Utkarsh Ankit

uankit@asu.edu | LinkedIn | Website | Google Scholar

Skills

Languages: Python, C++, JavaScript, SQL, Bash, Linux, PyTorch, JAX, ROS, [DSA and System Design]
Recommendation System: Collaborative Filtering, Content-Based Filtering, Hybrid approaches, Two-Tower Algorithm
Agentic Systems & LLMs: AI Agent Architecture, LangChain, LangGraph, MCP, RAG, Multimodal and Tools Integrations
AI & Machine Learning: (M)LLM Post-training & Fine-tuning, Prompt Engineering, LLM Orchestration, LLM APIs, HNSW
Databases and Search Relational and Non Relational, Graph Databases (Neo4j), Vector Databases (Pinecone), Elastic Search, RRF
Reinforcement Learning: PPO , SAC, TD3, Reward Function Design, Observation/Action Space Formulation, Imitation Learning
Infrastructure & MLOps: AWS, CI/CD, Distributed Systems, MVP-stage AI platforms, Computer architecture
Simulation & Environments: MuJoCo, NVIDIA Isaac Sim, MJCF/XML, Gymnasium Robotics, Sim-to-Real
Evaluation & Monitoring: Agent Evaluation Frameworks, OpenTelemetry, Shadow testing, A/B testing, NDCG, Reproducibility

Education

M.S. (Thesis Track) in CSE (Robotics & Artificial Intelligence) 2023 – 2025
Arizona State University (ASU)

B.Tech. in Computer Science & Engineering 2016 – 2020
Kalinga Institute of Industrial Technology

Experience

AI Research Engineer (Reinforcement Learning and Planning) 2024 – Present
Cognitive Robotics & Safe Autonomy (CRS) Lab, Arizona State University

- Formulated safety guarantees for Robust Constrained MDPs (RCMDPs) using PPI and CBFs in stochastic environments.
- Benchmarked Safe RL algorithms **(SAC, PPO, CPO)** in high-fidelity simulators (NVIDIA Isaac Sim, MuJoCo); reduced model's latency by **40%** by harnessing GPU acceleration through custom **CUDA kernels** for non-standard activation functions on HPC.
- Value iteration to optimize constraint satisfaction under stochastic transitions, specifically focusing on **Explainability in AI**.

AI Engineer (LLMs, RAGs and Agentic Systems) 2024 – 2025
Embodied Games Lab, Arizona State University

- Architected a **multimodal RAG agent** acting as a Socratic physics instructor; orchestrated LLMs to process real-time LiDAR motion data, speech transcripts, and text to generate personalized student guidance.
- Engineered **end-to-end Generative AI pipelines** on **AWS** to benchmark SOTA models (OpenAI's GPT-4o, Claude 3.5, Gemini 1.5); implemented local LLM inference for Llama 3.2 to **ensure low-latency responses**.
- Developed the Socratic Precision F-Score (SPF), a novel metric to penalize semantic false alarms and hallucinations, outperforming standard ROUGE, SBERT cosine similarity in pedagogical evaluation, utilising an **LLM-as-a-judge** framework.
- Optimized model on resource-constrained devices using Singular Value Decomposition (**SVD**) for model compression.

Software Engineer (Data Science and MLOps) 2020 – 2023
BusinessNext

- Built and productionized **ML models** for banking CRM workflows (**lead scoring**, **forecasting**, and customer insights), improving business accuracy by **18%** and increasing downstream lead conversion.
- Owned **end-to-end ML pipeline** components: data extraction, **feature engineering**, training, evaluation, and deployment handoff; added automated refresh/retraining triggers based on data and performance signals.
- Designed scalable feature pipelines on **MS SQL Server** and **Python**; optimized stored procedures, indexing, and query plans to reduce pipeline runtime by **30%**.
- Deployed model inference behind **REST services** consumed by CRM modules; improved response times and reduced data retrieval **latency** by **15%** via API contract optimization and caching patterns.
- Implemented model quality tracking and **A/B testing** pipelines to evaluate production feedback loops, translating experimental results into actionable product decisions and documented runbooks with customer teams.

Research and Engineering

Command & Data Handling (CDH) Engineering (MLOps) - ***(NASA L'Space)***

- Led data engineering and ETL pipelines for mission telemetry ingestion, processing, and analysis. Gained System Design experience.

Neural Machine Translation for Indian Languages (Transformer NN) - ***(IIT-BHU)***

- Engineered a Transformer-based NMT model with positional encoders, boosting translation accuracy by 15%.
- Integrated Monotonic Chunk-wise Attention, achieving a 3% improvement on supercomputing infrastructure.

Image Segmentation of Satellite & Drone Images (Computer Vision) - ***(ISRO)***

- Devised and fine-tuned a CNN-based segmentation pipeline to detect buildings in satellite imagery, achieving a 95% Dice Score.

Awards, Additional Experience & Certifications

Award: SCAI Outstanding Graduating MS Thesis Student (AI & Robotics) – *Arizona State University, Spring 2026*

Technical Blogger @Towards Data Science: 175+ subscribers, 2K+ views, blog featured on BuiltIn.com.

MLOps Certification: (Link) | **Guest Speaker**: RoboticsLab (Peru) (Link) | **Google Scholar**: 28 citations